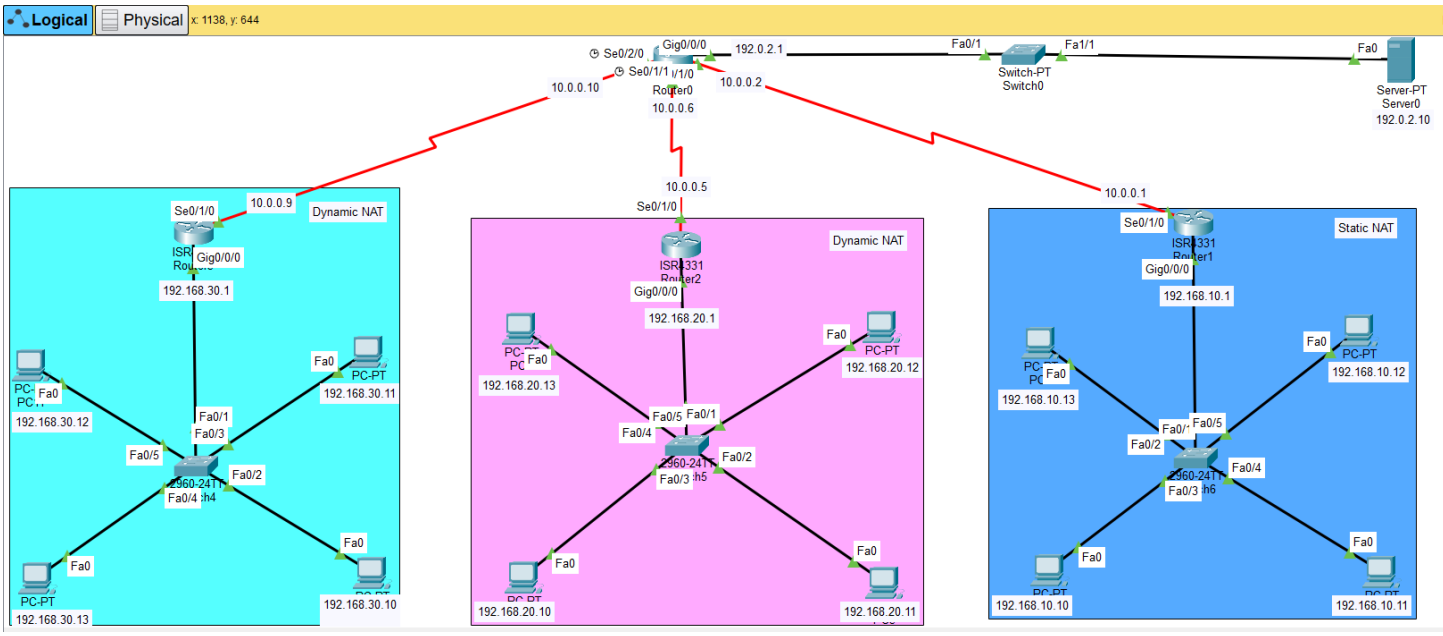
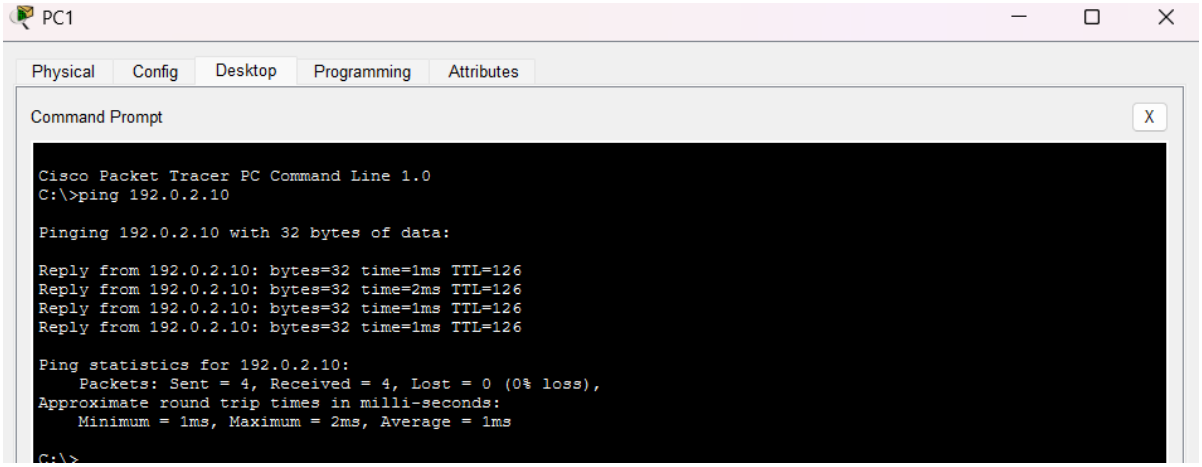
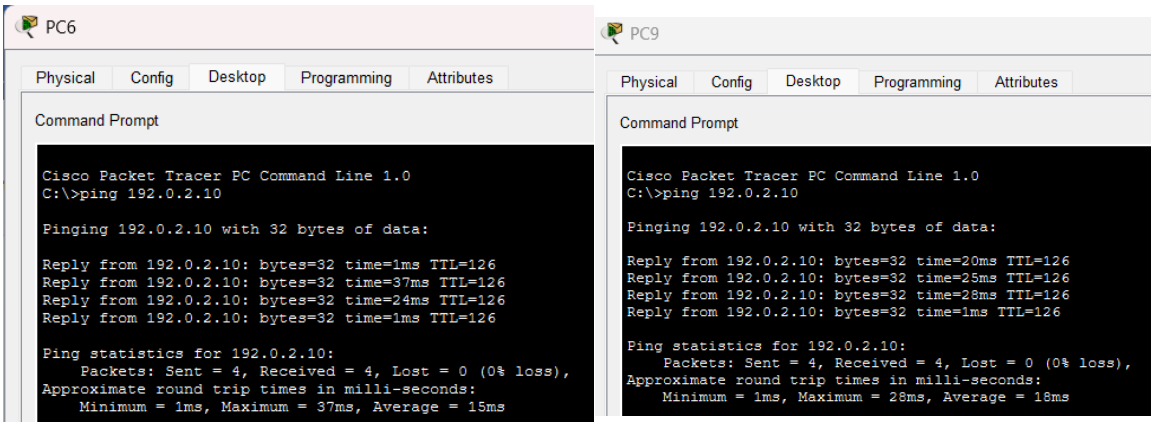


Verification Steps

1. From Branch1 PC (with static NAT), ping the internet server (192.0.2.10)



2. From PCs in Branch2 and Branch3, ping the same server to test dynamic NAT.



3. On R0, use: show Ip Nat translations, show Ip Nat statistics, ping, traceroute, debug Ip Nat

```
R0_Anup7985#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 203.0.113.5:8     192.168.30.11:8    192.0.2.10:8       192.0.2.10:8
icmp 203.0.113.5:9     192.168.30.11:9    192.0.2.10:9       192.0.2.10:9
icmp 203.0.113.6:3     192.168.20.10:3    192.0.2.10:3       192.0.2.10:3
---  203.0.113.1       192.168.10.10      ---                ---

R0_Anup7985#show ip nat statistics
Total translations: 2 (1 static, 1 dynamic, 1 extended)
Outside Interfaces: GigabitEthernet0/0/0
Inside Interfaces: Serial0/1/0 , Serial0/1/1 , Serial0/2/0
Hits: 10  Misses: 22
Expired translations: 11
Dynamic mappings:
-- Inside Source
access-list 1 pool DYNAMIC_POOL refCount 1
 pool DYNAMIC_POOL: netmask 255.255.255.0
       start 203.0.113.2 end 203.0.113.10
       type generic, total addresses 9 , allocated 0 (0%), misses 0
R0_Anup7985#ping 192.168.30.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/16/33 ms

R0_Anup7985#traceroute 192.168.30.10
Type escape sequence to abort.
Tracing the route to 192.168.30.10

  1   10.0.0.9        0 msec    1 msec    18 msec
  2   192.168.30.10   1 msec    1 msec    8 msec
R0_Anup7985#debug ip nat
IP NAT debugging is on
R0_Anup7985#
NAT: s=192.168.20.13->203.0.113.6, d=192.0.2.10 [5]

NAT*: s=192.0.2.10, d=203.0.113.6->192.168.20.13 [28]

NAT: s=192.168.20.13->203.0.113.6, d=192.0.2.10 [6]

NAT*: s=192.0.2.10, d=203.0.113.6->192.168.20.13 [29]

NAT: s=192.168.20.13->203.0.113.6, d=192.0.2.10 [7]

NAT*: s=192.0.2.10, d=203.0.113.6->192.168.20.13 [30]

NAT: s=192.168.20.13->203.0.113.6, d=192.0.2.10 [8]

NAT*: s=192.0.2.10, d=203.0.113.6->192.168.20.13 [31]
```

Network Design Summary The topology implements a star-hub architecture connecting three branch office subnets (192.168.10.0/24, 192.168.20.0/24, and 192.168.30.0/24) to a central hub router (Router0) via point-to-point serial links (10.0.0.0/30 subnets). Router0 connects to an external DMZ switch (Switch0) on Gig0/0/0 (192.0.2.1/24), enabling access to external server 192.0.2.10. Static routing ensures full inter-branch reachability.

NAT Configuration Details Router0 executes both Static and Dynamic NAT. Static NAT translates host 192.168.10.10 directly to public IP 203.0.113.1. Dynamic NAT uses an ACL permitting internal subnets 192.168.20.0/24 and 192.168.30.0/24 to dynamically map traffic across pool NATPOOL (203.0.113.2–203.0.113.10). All inside serial interfaces use Ip Nat inside, while Gig0/0/0 is assigned Ip Nat outside.